Importing the required libraries

```
import pandas as pd
```

Opening the csv file for cleaning and filling null values

```
df = pd.read_csv('/content/customer_shopping_behavior.csv'
```

```
df.head()
```

Show hidden output

```
df.info()
```

Show hidden output

```
df.describe(include = 'all')
```

Show hidden output

```
df.isnull().sum()
```

Show hidden output

Since, reviews column has null values so

```
df['Review Rating'] = df.groupby('Category')['Review Ratir
```

```
df.isnull().sum()
```

Show hidden output

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
```

```
df.columns
```

Show hidden output

```
df = df.rename(columns = {'purchase_amount_(usd)':'purchase_amount'})
```

```
df.columns
```

Show hidden output

Create a column age_group

```
labels = ['Young', 'Adult', 'Middle-aged', 'Senior']
df['Age_group'] = pd.qcut(df['age'], q = 4, labels = labels)
```

```
df[['age', 'Age_group']].head(10)
```

	age	Age_group
0	55	Middle-aged
1	19	Young
2	50	Middle-aged
3	21	Young
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young
8	26	Young
9	57	Middle-aged

Let's create another feature called purchase_frequency_days

```
frequency_mapping = {
    'Fortnightly': 14,
    'Monthly': 30,
    'Weekly': 7,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
df[['discount_applied', 'promo_code_used']].head(10)
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
np.True_
```

Since both columns are same then we can remove one column

```
df = df.drop('promo_code_used', axis = 1)
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
```

```
       'frequency_of_purchases', 'Age_group', 'purchase_frequency_days'],
      dtype='object')
```

```
df.to_csv('customer.csv', index = False)
```

```
from google.colab import files
files.download("customer.csv")
```

```

```

```
# Another way to load the table into postgresql

!pip install pandas sqlalchemy psycopg2-binary
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3
Requirement already satisfied: sqlalchemy in /usr/local/lib/python3.13/dist-packages (2
Requirement already satisfied: psycopg2-binary in /usr/local/lib/python3.13/dist-packag
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-package
Requirement already satisfied: typing-extensions>=4.6.0 in /usr/local/lib/python3.13/di
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (frc
```